

AKASH SARAVANAN

GEOSPATIAL ENGINEER

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PROFILE SUMMARY

Geospatial Engineer with 3 years 8 months of experience delivering GIS mapping, spatial analysis, and geodatabase management across infrastructure and land development projects. Specialized in QGIS, ArcGIS Pro/Online, and Python-based geospatial automation (GDAL, Rasterio, GeoPandas). Led QA/QC processes achieving 100% data integrity across cadastral and property tax geospatial projects, while mentoring a 2-person team. Focused on translating complex spatial data into accurate, decision-ready outputs that support infrastructure planning and land development goals.

TECHNICAL SKILLS

- GIS & Mapping Tools: ArcGIS Pro, ArcGIS Online, Field Maps, QGIS, AutoCAD Map 3D, Google Earth Pro, Global Mapper
- Data & Geodatabase Management: Geodatabase structuring, metadata documentation, folder structure/naming conventions, version control, vector/raster conversion, georeferencing, geoprocessing
- BIM – GIS Integration: Coordinating GIS/Spatial data with CAD & BIM drawings, leveraging civil engineering domain knowledge.
- Development & Automation: Python (GDAL, Rasterio, GeoPandas — geospatial workflow automation), Model Builder, FME, PostGIS, SQL
- Drone Data Processing: Pix4D mapper, Agisoft Metashape, Reality Capture, DJI Terra
- Spatial Analysis & Cartography: Multi-criteria Spatial analysis, cartographic map production, symbology and layout design
- Productivity: Microsoft 365 (Excel, PowerPoint, Word)

EXPERIENCE

FARMWISEAI PRIVATE LIMITED – GIS Analyst; Chennai, India

September 2025 – Present

- Techno-Economic Feasibility Studies (TEFR): Conducting comprehensive spatial and economic analysis for proposed Industrial Parks, evaluating land potential, infrastructure connectivity, and investment viability.
- Coordinated GIS datasets with CAD/BIM survey drawings for industrial park layouts, applying civil engineering knowledge to ensure spatial and design consistency across disciplines
- Site Suitability Modeling: Developed multi-criteria decision-making models to identify optimal locations for industrial setups, balancing technical constraints with environmental and logistical factors.
- Precision Drafting & Layout: Executing end-to-end AutoCAD FMB (Field Map Book) stitching, transforming raw survey data into complete, high-accuracy industrial layouts.
- Quality Assurance (QA/QC): Leading the quality control process for Property Tax Geospatial Projects using QGIS, ensuring 100% data integrity between physical spatial records and tax databases.
- Web GIS Platform Delivery: Contributed to the design and delivery of a GIS-based layout management platform (proof of concept) for CUMTA (Chennai Unified Metropolitan Transport Authority), covering 40+ CMDA layout projects — including project management, geodatabase/data pipeline design (auto-georeferencing of CAD/PDF layouts to GeoJSON), and coordination of the technical build.
- FPO Portal Contribution: Supporting ongoing development of an FPO (Farmer Producer Organization) platform — contributing to frontend UI improvements and independently building land-document retrieval workflows (Patta/Chitta, A-Register) to feed accurate cadastral data into the farmer database.
- GIS Process Automation: Streamlined daily GIS workflows by developing automated scripts and tools to reduce manual data entry and increase processing speed.
- Advanced Geospatial Research: Actively implementing GeoAI models to integrate Machine Learning with spatial data for predictive land-use analysis and automated feature extraction.
- Stakeholder Presentation: Translating complex technical suitability reports into visual, data-driven presentations for executive decision-making.
- Web GIS & Field Data Collection: Using ArcGIS Online for data visualization/dashboarding and Field Maps for field data collection, supporting site assessment and survey capture workflows.

I. OEG DEFENCE AND AEROSPACE PRIVATE LIMITED – GIS Analyst; Chennai, India

- Develop GIS datasets and topographic maps for engineering applications
- Produced cartographic outputs (site maps, layout plans, thematic maps) using QGIS for engineering review and client submission.
- Using ArcGIS Online and Field Maps for field data collection and visualization to support project data workflows.
- Ensured positional accuracy and consistency of map outputs through systematic QA/QC checks.

II. OPERATIONAL ENERGY GROUP INDIA LIMITED – GIS Analyst; Chennai, India

- Conducted volumetric analysis and topographic mapping for thermal power plant sites.
- Prepared land survey-based mapping deliverables to support infrastructure planning.
- Maintained mapping standards and symbology conventions across project deliverables.

III. ZUPPA OEG GEN5 TECHNOLOGIES PRIVATE LIMITED – GIS Analyst; Chennai, India

- Processed drone-captured imagery into georeferenced Orthomosaics and digital elevation models for site mapping.
- Generated topographic outputs from UAV survey data for engineering and client review.
- Supported geospatial data integration between drone-derived datasets and GIS platform (QGIS) for mapping deliverables.
- Provided mapping-related technical support and demonstration at client meetings.

SSG ENGINEERS – Site Engineer; Salem, India

December 2021 – December 2022

- Monitored daily construction activities to ensure adherence to project plans and safety standards.
- Coordinated with contractors and suppliers to maintain efficient workflows and timely material deliveries.
- Maintained accurate documentation of project progress inspections and site reports.

EDUCATION

PANIMALAR ENGINEERING COLLEGE – Chennai, INDIA

September 2020

Bachelor of Engineering, Civil Engineering; Cumulative GPA: 7.32/10.0

THIAGARAJAR POLYTECHNIC COLLEGE – Salem, INDIA

May 2017

Diploma, Civil Engineering; Cumulative Percentage: 86.74

PROJECTS

- **Site Suitability Analysis for Proposed Industrial Parks** Conducted multi-criteria spatial analysis to evaluate and identify optimal locations for newly proposed industrial parks. Integrating vector datasets (Land use, infrastructure proximity) and raster datasets (terrain, environmental layers) to generate suitability maps for data-driven site selection.
- **CUMTA Web GIS Platform (PoC)** Contributed to the design and development of a Web GIS proof-of-concept platform for visually managing CUMTA layout data across 40+ layout projects, coordinating technical delivery on a modern stack using AI-assisted development tools. Focused on translating cadastral and layout data into an accessible visual interface for project stakeholders.
- **FPO Portal** Map-first management cockpit for Farmer Producer Organizations, where a survey-level land map and a live MIS dashboard work as one instrument. Integrated real government data (land parcels, soil-health cards, fertilizer dealer stock) and built the village-drill, parcel-selection and machinery-rental experiences for a 323 member FPO in Tamil Nadu
- **Techno-Economic Feasibility Study for Industrial Park Development** Contributed geospatial inputs to techno-economic feasibility reports for proposed industrial parks, combining spatial analysis with infrastructure assessment. Produced GIS-based layouts and location intelligence that informed planning, investment viability, and regulatory compliance decisions.

- **Cadastral FMB Stitching & Large-Scale Land Parcel Mapping** Executed AutoCAD-based FMB stitching workflows for land parcels ranging from 200 to 2,000 acres across multiple project sites. Ensured spatial accuracy and boundary integrity for cadastral records, supporting land acquisition and legal documentation processes in large-scale infrastructure projects.
- **Photogrammetry Mapping & Volumetric Estimation** — Highways, Railways & River Beds Processed aerial survey data to generate DSMs, Orthomosaics, and Contour maps for National Highways, Railway corridors, and River bed projects. Performed volumetric estimation for coal stockpiles using point cloud and elevation data, delivering high-accuracy outputs critical for infrastructure planning and resource quantification.

KEY COMPETENCIES

- Photogrammetric Mapping
- Cadastral Mapping & Geospatial Data Quality Control
- GIS Project Planning, Execution & Delivery
- Analytical Thinking & Problem Solving
- Strong Communication & Cross-functional Collaboration

CERTIFICATIONS AND TRAINING

- Advanced GIS and Remote Sensing (ArcGIS Pro, QGIS)
- AutoCAD Certification